

CSCI 5942: AI Engineering

Fall 2026

Version: August 14, 2026

Description: This course covers the infrastructure of modern AI: the models, the data, and the systems that train and serve them. Generally, Heckman will be lecturing on the models (scaling laws, architectures, data curation, fine-tuning, post-training, evaluation, applications) and Zhao will be lecturing on the systems (MLOps, data infrastructure, distributed training, inference serving, agentic systems). By the end of the semester you will have trained, adapted, evaluated, quantized, and served models you built on infrastructure you have configured!

Instructors: Christoffer Heckman (ECES 130) and Mark Zhao (ECCR 1B26).

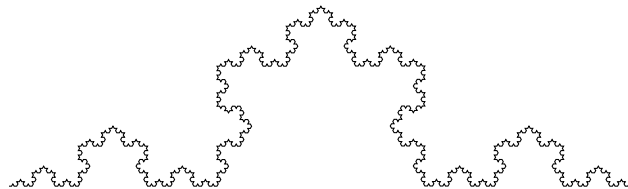
Teaching Assistant: Nolan Brady (Nolan.Brady@colorado.edu).

Meeting times: Tuesdays & Thursdays. 2:00–3:15, MBE 155.

Office Hours: Posted as an announcement in Slack.

Prerequisites: CSCI 4622, CSCI 4922.

Communication: The course website is the source of truth for the schedule and readings. Assignments live in the course GitHub organization with starter code; submissions go through Gradescope; announcements and discussion happen on the course Slack.



Structure of this Course

Lectures and readings: Each lecture has one required paper and possibly a short list of optional ones. Please spend some time with the required reading before class!

Participation (Round Robin): Discussion is Socratic and round robin: one or two students are “on deck” each meeting and lead the response to discussion questions. Being on deck is scheduled $\approx$ one lecture in advance. The questions rarely have a single right answer, and grading rewards engaged reasoning, not correctness.

Assignments: Seven graded assignments, roughly biweekly, released on GitHub with skeleton/starter code. The assignments are constructive; each produces an artifact (a trained model, an eval harness, a data pipeline) that later lectures and assignments build on.

A0: Environment setup and Google Cloud credit signup. A1 requires it.

- A1: A decoder-only Transformer from scratch, on Shakespeare, plus a small scaling study.
- A2: Evaluation harness and tracing (Opik) for the A1 model.
- A3: Spark, data lakes, and dataloaders.
- A4: Multi-system training.
- A5: Supervised fine-tuning and PEFT.
- A6: Inference optimizations.
- A7: RLAIIF.

Quizzes: Each assignment is followed by a short, closed-book, in-class quiz on the material of that assignment. If you did the assignment yourself, the quiz is straightforward. Quizzes are written by both instructors and account for 3 of each assignment's 7 points.

Late policy: Each assignment may be submitted up to two days late without penalty; beyond that, 25% per additional day. You only have *five* late days to use through the semester. Late days are calculated automatically based on submission time.

Mega Assignment (group project): A semester-long group build in place of an eighth assignment, on a topic you propose. Deliverables:

Concept document (5 pts): what you will build, on what data and compute, and how you will know it worked.

Pitch day (5 pts): a short presentation and defense of the plan.

Final submission (10 pts): code, README, and results, presented at the poster session on Thursday, December 3.

Final Exam: In person, written, during the scheduled final slot (Wednesday, December 9).

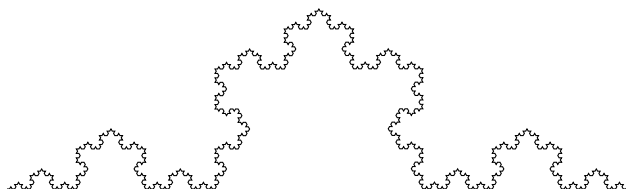
Grading:

- Assignments: 50 pts (A0 1 point, then 7 assignments $\times$ 7 pts; 4 for the assignment, 3 for its quiz)
- Participation (round robin): 10 pts
- Mega assignment: 20 pts (5 concept + 5 pitch + 10 final)
- Final exam: 20 pts

Computing: Every assignment runs on hardware you have or credits you are given: Google Cloud credits (signup in week 1), course allocations on Chameleon and NSF ACCESS (both with high-bandwidth-memory nodes), and the course Slurm cluster (4 $\times$ RTX 8000, 48 GB each) coming online at the end of August. Where a topic needs frontier-scale numbers, we read the papers of the people who paid for them. The mega assignment includes deciding, and defending, your own compute plan.

Lecture Schedule:

Date (T)	Topic	Lead	Date (Th)	Topic	Lead
Aug 25	Scaling Laws & Model Codeign	CH	Aug 20	Introduction (A0 out)	Both
Sep 1	ML Systems and MLOps	MZ	Aug 27	Model Architectures & Trade-offs (A1 out)	CH
Sep 8	Data Synthesis & Curation	CH	Sep 3	Model Evaluation (A2 out)	CH
Sep 15	Data Infrastructure II (A3 out)	MZ	Sep 10	Data Infrastructure I	MZ
Sep 22	Training Systems II	MZ	Sep 17	Training Systems I	MZ
Sep 29	Project Day	Both	Sep 24	Training Systems III (A4 out)	MZ
Oct 6	Inference Systems I (A5 out)	MZ	Oct 1	Fine-tuning & PEFT	CH
Oct 13	Inference Systems II	MZ	Oct 8	<i>Reading Day (campus)</i>	–
Oct 20	Model Optimizations for Inference (A6 out)	CH	Oct 15	Inference Systems III	MZ
Oct 27	Post-training I: RLHF & GRPO	CH	Oct 22	Model Quantization	CH
Nov 3	Agentic AI (A7 out)	MZ	Oct 29	Post-training II: RLVR	CH
Nov 10	Sustainability and ML	MZ	Nov 5	Prompt Engineering & Tools	CH
Nov 17	Applications: Vision at Scale	CH	Nov 12	Applications: Robotics	CH
Nov 24	<i>Fall break</i>	–	Nov 19	Applications: GeoAI	CH
Dec 1	Last Lecture	Both	Nov 26	<i>Fall break</i>	–
			Dec 3	Poster Session	Both
			Dec 9 (W)	Final Exam	–



Provost's Statements

Honor Code

All students enrolled in a CU-Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part of adhering to the Honor Code. All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution.

In this course, please respect the following specific policies:

- **AI Tools:** AI usage is permitted, treated as if the AI were a colleague: you may consult it, delegate to it, and learn from it, and you are responsible for your work, including its correctness and your understanding of it. The in-class quizzes and the in-person final exist so that understanding is checked in person. For the mega assignment, AI use is expected; document it as you would any collaborator.
- **Collaboration:** Study groups are allowed and encouraged. Each student submits their own assignment work and lists study-group members at the top of the submission.
- **Quizzes and exam:** Closed book, no devices, no AI.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. If you have a temporary illness, injury, or required medical isolation for which you require adjustment, please email me before class starts, ideally the day before. I cannot accommodate explanations after the fact for the purpose of allocating your participation. Your email should not include any details but merely state that you cannot attend due to medical reasons.

Accommodation for Religious Obligations

Campus policy requires faculty to provide reasonable accommodations for students who, because of religious obligations, have conflicts with scheduled exams, assignments, or required attendance. Please provide me notice of the need for a religious accommodation within the first two weeks of class.

Preferred Student Names and Pronouns

CU Boulder recognizes that students' legal information does not always align with how they identify. If you wish to have your preferred name and/or preferred pronouns appear on your instructors' class rosters and in Canvas, visit the Registrar's website for instructions.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering an inclusive and welcoming learning, working, and living environment. University policy prohibits protected-class discrimination and harassment, sexual misconduct, intimate partner abuse, stalking, and related retaliation. Faculty are required to inform the Office of Institutional Equity and Compliance (OIEC) when they are made aware of incidents related to these concerns.

Mental Health and Wellness

If you are struggling with personal stressors, mental health or substance use concerns that are impacting academic or daily life, please contact Counseling and Psychiatric Services (CAPS) at (303) 492-2277, 24/7.